NORTH BENGAL INTERNATIONAL UNIVERSITY

Lab Report : 1

TOPIC : ASSEMBLY LANGUAGE BASIC
COURSE CODE : CSE 0611-3322
COURSE TITLE : MICROPROCESSOR & ASSEMBLY LANGUAGE SESSIONAL.

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Task Number : 01

Task Title : Basic Assembly Language

Code:

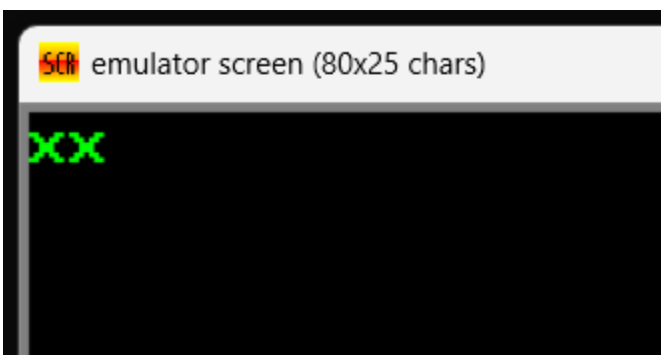
```
.model small
.stack 100h
.code
main proc
        mov ah, 1
        int 21h

        mov ah, 2
        mov dl, al
        int 21h

        exit:
        mov ah, 4ch
        int 21h

        main endp
end main
```

Output:



Theory

- **.model small:** Defines a small memory model in assembly language.
- **.Code and Data:** Code and data are stored in separate 64KB segments.
- **For Small Programs:** Suitable for programs where the code and data each fit within 64KB.
- **.stack 100h:** Sets the size of the stack.
- **100h:** Means 256 bytes of memory for the stack. Stores temporary data like function info and variables.

• **.data:** Marks the beginning of the data section in an assembly program. Used to define and store variables or constants that the program will use.

• **.code:** Marks the beginning of the code section in an assembly program. It tells the assembler that the following lines are executable instructions.

• **Execution:** The CPU reads and executes the instructions after this directive.

• **mov ah, 4ch:** Prepares the `AH` register with the value '4CH', which is the code for "terminate program" in DOS.

• **int 21h:** Calls the DOS interrupt service. With 'AH' set to '4CH', this instruction ends the program and returns control to the operating system.

• **mov ah, 1:** Prepares to read a single character from the keyboard.

• **int 21h:** Executes the DOS interrupt to read the character and store it in the `AL` register.

• **mov ah, 2:** Prepares to output a character to the screen.

• **mov dl, al:** Moves the character from the `AL` register to the `DL` register, which is used for output.

• **int 21h:** Executes the DOS interrupt to display the character stored in `DL` on the screen.

• **exit:** This is just a label marking the next part of the program. It doesn't do anything by itself but serves as a reference point in the code.

Conclusion: In this lab, we gained basic knowledge and learned more about the 8086 processor. We also applied and observed the basic codes.